

Integrated Working Edition

WORKING EDITION: this volume currently contains the frozen practice questions from ST01-ST12. The master audit identifies additional clear permutation-owned ExamSIDE rows that must be added before this volume is labelled complete.

CONTENTS

ST01	Counting Engine
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ST11 Forbidden Strings / Inclusion-Exclusion

ST12 Hybrid Permutation Structures

EXAMSIDE PRACTICE

Source-linked PYQs -> concept backlinks -> H1/H2/H3 -> Appendix A solutions -> coverage audit

PRACTICE RULE

Try the question without a helper first.

Use H1 for recognition, H2 for a visual model, H3 only for setup rescue.

Solutions remain in Appendix A so question pages do not reveal the answer.

This PDF audits local ST01 coverage; chapter-wide completeness remains provisional.

PRACTICE MAP + RELEASE GATE

Four tightly scoped ExamSIDE questions

MAP

Question map

Q1

JEE Main 2018 | 15 Apr Morning

C3 Repetition

Q2

JEE Main 2026 | 28 Jan Evening

C4/C6 Ordered assignment

Q3

JEE Main 2023 | 24 Jan Evening

C6 Hidden permutation

Q4

JEE Main 2024 | 8 Apr Morning

C6 Hidden permutation

SCOPE

Anti-drift

Only ST01 counting-engine questions are included. Questions whose real difficulty is blocks, gaps, repeated objects, digit restrictions, dictionary rank or circular seating are deferred.

Helper and integrity rules

H1

recognition cue only

H2

visual / structural helper

H3

setup skeleton; no final arithmetic

LINK

source + concept backlink

SOLUTION

Appendix A only on question attempt pages

READABILITY

Typography gate

v5 uses larger question/helper text and one solution per page. Dense solution spreads were split instead of shrinking the font.

Q1

D1

BACK TO C3 Repetition

OPEN ORIGINAL

Using only the digits 2, 5 and 7, n -position strings/numbers are formed with repetition allowed. Find the least n for which at least 900 distinct outcomes are possible.

MISCONCEPTION

Watch for this trap

Distinct positions $\rightarrow$ wrongly forbids repetition

Working

Helpers - use only if needed

H1

Recognition cue

Repetition is allowed. Does any position lose a digit after you use it?

H2



H3

Structural setup

Write the family 3^n and compare successive powers with 900. Stop before the final arithmetic if you can continue independently.

Solution $\rightarrow$ Appendix A

Q2

D2

[BACK TO C4/C6 Ordered assignment](#)[OPEN ORIGINAL](#)

Three distinct passengers enter a lift. The lift can stop at 7 permitted floors, and the passengers must exit at three different floors. Count the possible exit assignments.

MISCONCEPTION**Watch for this trap**

Choose floors only -> misses passenger assignment

Working**Helpers - use only if needed**

H1

Recognition cue

Choosing only the set of floors is incomplete. What else is recorded?

H2

P1

7

x

P2

6

x

P3

5

H3

Structural setup

Structural skeleton: $7 \times 6 \times 5$. Do not evaluate until every factor has a meaning.

Solution -> Appendix A

Q3

D3

BACK TO C6 Hidden permutation

OPEN ORIGINAL

Count 5×5 zero-one matrices in which every row has exactly one 1 and every column also has exactly one 1.

MISCONCEPTION

Watch for this trap

Treat rows independently $\rightarrow 5^5$ and reuse columns

Working

Helpers - use only if needed

H1

Recognition cue

Treat each labelled row as a slot choosing the column that contains its 1.

H2

0	1	0	0	0
0	0	0	1	0
1	0	0	0	0
0	0	0	0	1
0	0	1	0	0

Once a column is used, the column-sum condition prevents reuse.

H3

Structural setup

Row-choice skeleton: $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$. Recognise the factorial before evaluating.

Solution $\rightarrow$ Appendix A

Q4

D3

[BACK TO C6 Hidden permutation](#)[OPEN ORIGINAL](#)

Let A be the set of prime factors of 2310 and $f(x) = \text{floor}(\log_2(x^2 + \text{floor}(x^3/5)))$. After determining the range of f on A , count the one-to-one functions from A to that range.

MISCONCEPTION**Watch for this trap**

Count arbitrary functions -> allows repeated outputs

Working

Helpers - use only if needed

H1

Separate the tasks

First determine the domain size and the number of distinct range values.

H2

Think of each labelled input as needing one distinct output. Each used output leaves the pool.

 x_1

5

x

 x_2

4

x

 $x_3 \dots$

3

H3

Structural setup

With five labelled inputs and five distinct outputs, the one-to-one assignment has a $5!$ structure.

Solution -> Appendix A

AFTER THE FOUR QUESTIONS: DIAGNOSE YOUR ERROR TYPE

Transferable recognition matters more than four answers

DIAG

Error categories

Recognition

R

I did not identify the ordered/repeated-slot structure.

Setup

S

I saw the concept but wrote the wrong slot counts.

Misconception

M

I used a tempting wrong model such as choose-only or arbitrary functions.

Arithmetic

The model was correct, execution failed

ACTION

Repair before progression

If Recognition, Setup or Misconception occurs twice or more, revisit the linked Study Guide concept before moving on.

Question-specific misconception check

Q1

Distinct positions -> wrongly forbids repetition

Q2

Choose floors only -> misses passenger assignment

Q3

Treat rows independently -> 5^5 and reuse columns

Q4

Count arbitrary functions -> allows repeated outputs

REFLECT

Metacognition

Which wrong model felt tempting even when you eventually solved the question? That is the misconception to rehearse deliberately.

A

Q1 | Answer: 7

Every position has 3 choices because reuse is allowed. Count = 3^n . Since $3^6 = 729 < 900$ and $3^7 = 2187 \geq 900$, the least n is 7.



$$3^6 = 729 < 900 \quad \text{and} \quad 3^7 = 2187 \geq 900$$

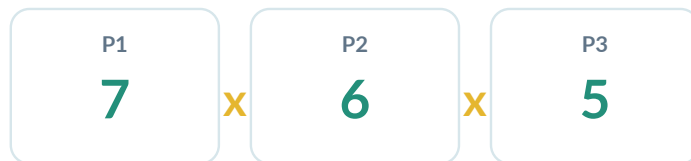
MISCONCEPTION REPAIR Pool stays constant

Repetition allowed means the pool stays at 3 for every position. Distinct positions do not force distinct digits.

A

Q2 | Answer: 210

Passenger 1 has 7 permitted floors, Passenger 2 then has 6 unused floors, and Passenger 3 has 5. Thus the ordered assignment count is $7 \times 6 \times 5 = 210$.



$$7 \times 6 \times 5 = 210$$

MISCONCEPTION REPORT Choose-only undercounts

7C3 selects floors but does not record the passenger-to-floor assignment.

A

Q3 | Answer: 120

Row 1 chooses one of 5 columns for its 1. That column cannot be used again because every column must also sum to 1. The row-wise choices are 5,4,3,2,1, giving $5! = 120$.

0	1	0	0	0
0	0	0	1	0
1	0	0	0	0
0	0	0	0	1
0	0	1	0	0

$$5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$$

MISCONCEPTION REPEATED Rows are not independent

5^5 lets different rows reuse the same column, violating the column-sum condition.

A

Q4 | Answer: 120

$A = \{2, 3, 5, 7, 11\}$. Evaluating f gives five distinct outputs $\{2, 3, 5, 6, 8\}$. A one-to-one map from a 5-element labelled domain to a 5-element range uses every output exactly once, so there are $5! = 120$.



$$5! = 120$$

MISCONCEPTION REPAIR: Injection -> permutation

5^5 counts arbitrary functions and permits repeated outputs. One-to-one means the output pool shrinks.

COVERAGE AUDIT | ST01

Local subtopic audit passes; chapter-wide audit remains provisional

AUDIT

Required ST01 rows currently identified

Q1

2018 n-digit 2,5,7

C3

Q2

2026 lift, distinct floors

C4/C6

Q3

2023 permutation matrix

C6

Q4

2024 one-to-one function

C6

RECONCILIATION ST01 local PASS

Required 4 | Included 4 | Source links 4 | Concept links 4 | Solutions 4 |
H-policy failures 0 | unresolved local rows 0

Deferred / not owned by ST01



fixed digit position / range

Number formation



repeated letters

Repeated objects



together / not together

Block + complement



no two together

Gap method



dictionary ordering

Lexicographic rank

CHAPTER STATUS Honest audit boundary

The full source corpus must be frozen and classified before chapter-wide completeness is claimed. Status: PROVISIONAL.

RELEASE CHECKLIST | ST01 v5

Coverage + pedagogy + readability

GATE

Pedagogy + readability gate

PASS real-life entry present

PASS meaning before notation

PASS concept helper / representation

PASS misconception + repair

PASS JEE trigger language

PASS transfer + retrieval

PASS body text legible at print scale

Practice integrity gate

PASS 4/4 source links

PASS 4/4 concept backlinks

PASS difficulty assigned 4/4

PASS misconception tags 4/4

PASS H-policy PASS

PASS 4/4 Appendix A solutions

PASS no final answers on question pages

FINAL STAMP

Release status

ST01-v5 | Required 4/4 | Source 4/4 | Concept 4/4 | Solutions 4/4 | H-policy PASS | Pedagogy PASS | Readability PASS | Chapter audit: PROVISIONAL

EXAMSIDE PRACTICE

Repeated Objects - source-linked PYQs with progressive helpers

HELPER POLICY

H1 = recognition cue; H2 = visual/concept model; H3 = structural skeleton only.

Final answers and full solutions are withheld from question pages and placed in Appendix A.

This local ST02 slice contains 4 required questions; chapter-wide audit remains provisional.

HOW TO USE THIS COMPANION

Attempt -> request minimum helper -> diagnose -> Appendix A

MAP

Question map

Q1

Prescribed repeated sequence

C3/C4 | D2

Q2

Choose repeated digits then arrange

C5 | D3

Q3

Two alike + two distinct

C5 | D3

Q4

Sum of repeated-digit numbers

C6 | D4

Self-diagnosis tags

IDENTICAL_SWAP_OVERCOUNT

counted fake labels

WRONG_MULTIPLICITY_DIVISOR

divided by wrong factorials

INVENTORY_STAGE_MISSED

arranged before choosing types

POSITION_FREQUENCY_ERROR

miscounted place contribution

RULE

Practice integrity

Do not open H2/H3 automatically. Use the smallest helper that restores the model; otherwise the practice stops measuring recognition.

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3/C4

IDENTICAL_SWAP_OVERCOUNT

D2

Question

QUESTION

JEE Main 2025 - 2 Apr Morning

A 10-term sequence uses only 0, 1 and 2, with exactly five 1s and exactly three 2s. How many such sequences are possible?

H1

Recognition cue

First infer the number of 0s. Then ask whether the five 1s are distinguishable from one another.

H2

Visual/concept helper

Think of 10 labelled positions. Choose which positions contain 1s, then which remaining positions contain 2s; the rest are 0s.

Your working

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C5

INVENTORY_STAGE_MISSED

D3

Question

QUESTION

JEE Main 2026 - 28 Jan Morning

Let $S=\{1,2,\dots,9\}$. Let x count 9-digit numbers in which exactly one digit is repeated twice and seven other digits occur once. Let y count 9-digit numbers in which exactly two digits are repeated twice and five other digits occur once. Determine $x:y$.

H1

Recognition cue

Do not start with 9!. First write the multiplicity patterns for x and y .

H2

Visual/concept helper

x has pattern $[2,1,1,1,1,1,1,1]$. y has pattern $[2,2,1,1,1,1,1]$. For each, choose which digit-types fill the repeated roles before arranging.

H3

Structural skeleton

$x = (\text{choose repeated digit})(\text{choose remaining singletons}) \times 9!/2!$. $y = (\text{choose two repeated digits})(\text{choose singletons}) \times 9!/(2!2!)$. Stop before simplifying.

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C5

INVENTORY_STAGE_MISSED

D3

Question

QUESTION

JEE Main 2020 - 5 Sep Morning

From the letters of SYLLABUS, form 4-letter words in which exactly one pair of letters is alike and the other two letters are distinct from each other and from the repeated letter. How many words are possible?

H1

Recognition cue

Which letters in SYLLABUS are actually available as a repeated pair?

H2

Visual/concept helper

Inventory pattern = $[2,1,1]$. Choose the repeated type from the letters that occur at least twice; then choose two distinct singleton types from the remaining distinct letter-types.

H3

Structural skeleton

(choices for repeated type) $\times$ (choices for two other distinct letters) $\times 4!/2!$.

Q4

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C6

POSITION_FREQUENCY_ERROR

D4

Question

QUESTION

JEE Main 2023 - 10 Apr Evening

Find the sum of all distinct 4-digit numbers formed using all the digits 1, 2, 2 and 3.

H1

Recognition cue

Avoid listing the numbers. Focus on one place value and count how often each digit appears there.

H2

Visual/concept helper

Fix the units place. With 1 fixed: arrange {2,2,3}. With 3 fixed: arrange {1,2,2}. With a 2 fixed: arrange {1,2,3}. Then use symmetry across thousands, hundreds, tens and units.

H3

Structural skeleton

Compute the total digit-contribution to one position, then multiply by 1111. Keep the repeated 2 multiplicity explicit when counting frequencies.

DIAGNOSTIC CHECKPOINT

Classify the error before reading a solution

DIAG

Q1

Did you use $10!$ without dividing for identical symbols?

Q2

Did you arrange before choosing which digits repeat?

Q3

Did you treat every letter as eligible for the repeated pair?

Q4

Did you assume every digit value has equal frequency in each place?

RETRY RULE

Repair before reveal

If the error is structural, return to the linked concept page and redo the question from a blank setup. Reading Appendix A immediately may fix the answer without repairing recognition.

LOCAL COVERAGE RECONCILIATION

Self-check against the frozen ST02 slice

AUDIT

Required rows

2025 sequence C3/C4 Q1 **PASS**

2026 repeated digits C5 Q2 **PASS**

2020 SYLLABUS C5 Q3 **PASS**

2023 repeated-digit sum C6 Q4 **PASS**

Deferred examples

BHBJO 50th word **DEFER -> Dictionary rank**

UDAYPUR rank **DEFER -> Dictionary rank**

16 colored cubes separated **DEFER -> Gap method**

INDEPENDENCE vowels together **DEFER -> Block + repeated objects**

RECONCILE

Local ST02 status: **PASS**

Required: 4
Included: 4/4
Source links: 4/4
Concept links: 4/4
Appendix solutions: 4/4
H-policy failures: 0

CHAPTER STATUS

Audit boundary

PROVISIONAL. This is a locally frozen ST02 slice, not proof that all 220 ExamSIDE P&C; rows have been classified. Chapter COMPLETE requires UNCLASSIFIED = 0.

A1

SOLUTION

Reasoning

The sequence has five 1s, three 2s and therefore two 0s. These are ten positions containing a multiset with multiplicities 5,3,2. Hence the count is $10!/(5!3!2!) = 2520$.

ANSWER

Final result

2520

A2

SOLUTION

Reasoning

For x: choose the repeated digit in 9 ways, choose 7 singleton digits from the remaining 8 in $C(8,7)$ ways, then arrange the multiset in $9!/2!$ ways. Thus $x = 9 \cdot C(8,7) \cdot 9!/2!$. For y: choose the two repeated digits in $C(9,2)$ ways, choose 5 singleton digits from the remaining 7 in $C(7,5)$ ways, then arrange in $9!/(2!2!)$ ways. The ratio simplifies to $x:y = 4:21$.

ANSWER

Final result

4:21

A3

SOLUTION

Reasoning

In SYLLABUS, only S and L are available as a repeated pair: 2 choices. After choosing the repeated type, there are 5 remaining distinct letter-types from which 2 different singleton letters are chosen: $C(5,2)$. For each inventory A,A,B,C there are $4!/2!$ arrangements. Total = $2 * C(5,2) * 4!/2! = 240$.

ANSWER

Final result

240

A4

SOLUTION

Reasoning

Use place-value contribution. In any fixed place: digit 1 occurs in $3!/2! = 3$ arrangements; digit 3 also occurs in 3 arrangements; digit 2 occurs in $3! = 6$ arrangements (counting visible numbers with a 2 fixed in that place). Total digit contribution to one place is $1*3 + 2*6 + 3*3 = 24$. By symmetry the same contribution occurs in every decimal place, so the total is $24*(1000+100+10+1) = 26664$.

ANSWER

Final result

26664

EXAMSIDE PRACTICE

Positional Restrictions - source-linked PYQs with progressive helpers

HELPER POLICY

H1 = recognition cue; H2 = structural/visual model; H3 = setup skeleton only.

Final answers and full solutions stay out of question pages and appear only in Appendix A.

Local ST03 slice contains 5 required questions; chapter-wide audit remains provisional.

HOW TO USE THIS COMPANION

Attempt -> minimum helper -> diagnose -> Appendix A

MAP

Question map

Q1

Fixed first/fourth letters

C2/C6 | D3

Q2

Dictionary locked in middle

C2 | D2

Q3

Odd digits in even places

C3/C4/C6 | D4

Q4

Even digits only even places

C3/C4/C6 | D3

Q5

Vowels in alphabetical order

C5 | D2

Self-diagnosis tags

FORMULA_FIRST_RESTRICTION_HIDDEN

counted before marking slots

FIXED_ITEM_DOUBLE_COUNT

rearranged a fixed object

POSITION_SET_NOT_CHOSEN

missed allowed-slot choice

ONLY_VS_EXACTLY

misread logical wording

RELATIVE_ORDER_SYMMETRY_MISUSED

overcased order condition

RULE

Practice integrity

Open the smallest helper that restores the model. H2/H3 are not rewards; they are repair tools for a failed representation.

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C2/C6

FORMULA_FIRST_RESTRICTION_HIDDEN

D3

Question

QUESTION

JEE Main 2016 - 9 Apr Morning

Using letters from MEDITERRANEAN, form four-letter words in which the first letter is R and the fourth letter is E. Repetition is limited by the available copies in the source word. How many distinct words are possible?

H1

Recognition cue

Lock R in position 1 and E in position 4 first. Only positions 2 and 3 remain to be filled.

H2

Visual/concept helper

After consuming one R and one E, remaining letter-types are M, E, D, I, T, R, A, N. Count ordered pairs of different types, then add the allowed same-letter pairs.

Your working

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C2

FIXED_ITEM_DOUBLE_COUNT

D2

Question

QUESTION

JEE Main 2018 - Offline

Choose 4 novels from 6 distinct novels and 1 dictionary from 3 distinct dictionaries, then arrange the 5 selected books on a shelf so that the dictionary is always in the middle. How many arrangements are possible?

H1

Recognition cue

Separate selection from placement: choose the books, lock the chosen dictionary in the centre, then arrange only the four novels.

H2

Visual/concept helper

Shelf skeleton: [_][_][D][_][_]. The centre no longer participates in the free factorial.

Your working

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3/C4/C6

POSITION_SET_NOT_CHOSEN

D4

Question

QUESTION

JEE Main 2019 - 8 Apr Morning

All 9-digit numbers are formed using 1,1,2,2,2,2,3,4,4 exactly once each. Count those numbers in which the odd digits occupy even-numbered positions.

H1

Recognition cue

There are 3 odd digits but 4 even-numbered positions (2,4,6,8). So the allowed-slot set is larger than the restricted group.

H2

Visual/concept helper

Choose which 3 of the 4 even positions hold 1,1,3. Arrange that odd multiset there. Fill all 6 remaining positions with 2,2,2,2,4,4.

H3

Structural skeleton

Count = $C(4,3) \times [3!/2!] \times [6!/(4!2!)]$. Stop before multiplying.

Q4

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3/C4/C6

ONLY_VS_EXACTLY

D3

Question

QUESTION

JEE Main 2023 - 24 Jan Morning

Using all digits of 123412341, form 9-digit numbers such that the even digits occupy only even-numbered positions. How many such numbers are possible?

H1

Recognition cue

Count the even digits and the even-numbered positions before interpreting “only”.

H2

Visual/concept helper

Even multiset = $\{2,2,4,4\}$; even positions = $\{2,4,6,8\}$. These four slots are therefore forced. Arrange the even multiset there and the odd multiset $\{1,1,1,3,3\}$ in the five remaining slots.

Your working

Q5

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C5

RELATIVE_ORDER_SYMMETRY_MISUSED

D2

Question

QUESTION

AIEEE 2004

Arrange all six distinct letters of GARDEN so that the vowels appear in alphabetical relative order. How many arrangements satisfy the condition?

H1

Recognition cue

The vowels are A and E. No exact positions are prescribed; only A must occur before E.

H2

Visual/concept helper

Pair every arrangement with its partner obtained by swapping only A and E. Exactly one member of each pair has A before E.

Your working

DIAGNOSTIC CHECKPOINT

Name the structural error before opening Appendix A

DIAG

Q1

Did you count middle slots before consuming the fixed R and E?

Q2

Did your final factorial still move the dictionary?

Q3

Did you forget to choose 3 of the 4 even positions?

Q4

Did you misread “only” before comparing counts?

Q5

Did you create exact-position cases instead of relative-order symmetry?

RETRY RULE

Repair before reveal

If the error is structural, return to the linked Study Guide concept and rebuild the slot map from scratch. Do not use Appendix A as the first repair.

LOCAL COVERAGE RECONCILIATION

Self-check against the frozen ST03 slice

AUDIT

Required rows

2016 R/E fixed C2/C6 Q1 **PASS**

2018 dictionary middle C2 Q2 **PASS**

2019 odd -> even places C3/C4/C6 Q3 **PASS**

2023 even only even C3/C4/C6 Q4 **PASS**

2004 GARDEN order C5 Q5 **PASS**

Deferred / excluded rows

2025 >50000 first+last condition **DEFER -> Digit formation**

2026 UDAYPUR rank **DEFER -> Dictionary rank**

2025 DAUGHTER vowels not together **DEFER -> Blocks/Complement**

2017 round-table adjacency **DEFER -> Circular**

AIEEE 2009 dictionary-middle duplicate **EXCLUDE -> duplicate fingerprint**

RECONCILE

Local ST03 status: **PASS**

Required: 5
Included: 5/5
Source links: 5/5
Concept links: 5/5
Appendix solutions: 5/5
H-policy failures: 0

CHAPTER STATUS

Audit boundary

PROVISIONAL. This locally frozen ST03 slice does not prove all 220 P&C; rows are classified. Chapter COMPLETE still requires UNCLASSIFIED = 0.

A1

SOLUTION

Reasoning

Fix R first and E fourth. After consuming one R and one E, the remaining letter-types are M,E,D,I,T,R,A,N. For the two middle positions: if the letters are different, there are $8 \times 7 = 56$ ordered pairs. A same-letter pair is possible only for E, A or N, because each still has at least two available copies: 3 more words. Total = $56 + 3 = 59$.

ANSWER

Final result

59

A2

SOLUTION

Reasoning

Choose 4 novels from 6 in $C(6,4)=15$ ways and choose 1 dictionary from 3 in 3 ways. The dictionary is fixed in the middle, so only the four selected novels are permuted in the remaining positions: $4!$ ways. Total = $15 \times 3 \times 24 = 1080$.

ANSWER

Final result

1080

A3

SOLUTION

Reasoning

Odd digits are 1,1,3 (3 objects). Even-numbered positions among 9 places are 2,4,6,8 (4 slots). Choose 3 of these 4 slots: $C(4,3)=4$. Arrange 1,1,3 there in $3!/2!=3$ ways. Fill the remaining 6 positions with 2,2,2,2,4,4 in $6!/(4!2!)=15$ ways. Total = $4 \times 3 \times 15 = 180$.

ANSWER

Final result

180

A4

SOLUTION

Reasoning

The even digits are 2,2,4,4: exactly 4 digits. The even-numbered positions are also exactly 4, so these slots are forced. Arrange the even multiset in $4!/(2!2!)=6$ ways. The remaining odd digits are 1,1,1,3,3, arranged in the five odd positions in $5!/(3!2!)=10$ ways. Total = $6 \times 10 = 60$.

ANSWER

Final result

60

A5

SOLUTION

Reasoning

All six letters of GARDEN are distinct, so there are $6!$ total arrangements. The only vowel-order possibilities are A before E and E before A. Swapping A and E pairs these possibilities one-to-one, so exactly half of all arrangements have the vowels in alphabetical order. Count = $6!/2 = 360$.

ANSWER

Final result

360

EXAMSIDE PRACTICE

Together / Block Method - source-linked PYQs with progressive helpers

HELPER POLICY

H1 = recognition cue; H2 = structural/visual model; H3 = setup skeleton only.

Question wording is paraphrased; buttons open the original ExamSIDE source.

Local ST04 slice has 3 required rows; negative-together and gap rows are deliberately deferred.

HOW TO USE THIS COMPANION

Attempt -> minimum helper -> diagnose -> Appendix A

MAP

Question map

Q1

INDEPENDENCE vowels together

C3 | D3

Q2

ASSASSINATION vowels together

C3 | D4

Q3

Girls block + boys block

C4/C6 | D4

Self-diagnosis tags

INNER_ORDER_FORGOTTEN

block treated as rigid

OUTER_MULTSET_WRONG

repeats outside mishandled

INNER_MULTISET_WRONG

repeats inside mishandled

MULTIPLE_BLOCKS_MERGED

independent groups glued together

PRIMARY_OWNER_DRIFT

mixed problem assigned to wrong engine

RULE

Practice integrity

Open the smallest helper that restores the model. H2/H3 should repair the block representation, not substitute for attempting the problem.

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3

INNER_ORDER_FORGOTTEN

D3

Question

QUESTION

JEE Main 2023 - 8 Apr Morning

Using every letter of INDEPENDENCE exactly as available, count the distinct arrangements in which all vowels occupy one consecutive block.

H1

Recognition cue

Glue I,E,E,E into one vowel block V. Keep consonant multiplicities visible outside V.

H2

Visual/concept helper

Outer inventory: V,N,N,N,D,D,P,C. Inner inventory: I,E,E,E,E. Count each level as a multiset arrangement.

Your working

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3

OUTER_MULTSET_WRONG

D4

Question

QUESTION

JEE Main 2023 - 1 Feb Morning

Using all letters of ASSASSINATION exactly as available, count the distinct arrangements in which all vowels occur consecutively as one block.

H1

Recognition cue

First split the inventory into vowels A,A,A,I,I,O and consonants S,S,S,S,N,N,T.

H2

Visual/concept helper

Glue all six vowels into V. Outer inventory: V,S,S,S,S,N,N,T. Inner inventory: A,A,A,I,I,O.

H3

Structural skeleton

Count = $[8!/(4!2!)] \times [6!/(3!2!)]$. Stop before evaluating.

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C4/C6

PRIMARY_OWNER_DRIFT

D4

Question

QUESTION

JEE Main 2025 - 22 Jan Evening

Three girls and four boys form a queue. All girls must form one block and all boys another block. Inside the boys block, two specified boys B1 and B2 must not be adjacent. How many queues satisfy all conditions?

H1

Recognition cue

At the outer level there are exactly two objects: G-block and B-block. Count their order first.

H2

Visual/concept helper

Girls inside: $3!$ orders. Boys inside: begin with $4!$ total orders, then remove boy-orders where B1 and B2 form one adjacent pair.

H3

Structural skeleton

Skeleton = $2! \times 3! \times [4! - 2 \times 3!]$. Stop before arithmetic.

DIAGNOSTIC CHECKPOINT

Name the structural error before opening Appendix A

DIAG

Q1

Did you count repeated N/D outside the block and repeated E inside separately?

Q2

Did you build two inventories before writing factorials?

Q3

Did you keep two outer blocks and apply the nonadjacency only inside the boys block?

RETRY RULE

Repair before reveal

If the error is structural, return to the linked Study Guide concept and redraw the outer inventory and inner inventory. Do not use Appendix A as the first repair.

LOCAL COVERAGE RECONCILIATION

Self-check against the frozen ST04 slice

AUDIT

Required rows

2023 INDEPENDENCE C3 Q1 **PASS**

2023 ASSASSINATION C3 Q2 **PASS**

2025 two blocks + B1/B2 C4/C6 Q3 **PASS**

Deferred rows

2025 DAUGHTER vowels never together **DEFER -> ST05 complement**

2026 queue all girls not together **DEFER -> ST05 complement**

2025 all boys OR no two boys **DEFER -> mixed ST04/ST06**

2023 C/S do not come together **DEFER -> ST05 complement**

2021 VOWELS consonants never together **DEFER -> ST05 complement**

RECONCILE

Local ST04 status: **PASS**

Required: 3
Included: 3/3
Source links: 3/3
Concept links: 3/3
Appendix solutions: 3/3
H-policy failures: 0

CHAPTER STATUS

Audit boundary

PROVISIONAL. This local ST04 slice is reconciled, but chapter **COMPLETE** still requires every one of the 220 P&C; rows to be classified.

A1

SOLUTION

Reasoning

INDEPENDENCE has vowels I,E,E,E,E and consonants N,N,N,D,D,P,C. Glue the vowels into one block V. The 8 outer objects V,N,N,N,D,D,P,C can be arranged in $8!/(3!2!)$ ways. Inside V, the vowels can be arranged in $5!/4!$ ways. Hence the total is $[8!/(3!2!)] \times [5!/4!] = 16800$.

ANSWER

Final result

16800

A2

SOLUTION

Reasoning

ASSASSINATION has vowels A,A,A,I,I,O and consonants S,S,S,S,N,N,T. Glue the six vowels into one block V. The 8 outer objects V,S,S,S,S,N,N,T can be arranged in $8!/(4!2!)$ ways. Inside V, the vowels can be arranged in $6!/(3!2!)$ ways. Therefore the count is $[8!/(4!2!)] \times [6!/(3!2!)] = 50400$.

ANSWER

Final result

50400

A3

SOLUTION

Reasoning

Treat the three girls as one block G and the four boys as one block B. The two blocks can be ordered in $2!$ ways. Girls have $3!$ internal orders. For the boys, there are $4!$ total orders. If B1 and B2 are adjacent, treat them as one unit: the three resulting units can be arranged in $3!$ ways and B1,B2 can swap in 2 ways, giving $2 \times 3!$ adjacent orders. So valid boy orders are $4! - 2 \times 3! = 12$. Total = $2! \times 3! \times 12 = 144$.

ANSWER

Final result

144

EXAMSIDE PRACTICE

Not Together / Complement - source-linked PYQs with progressive helpers

HELPER POLICY

H1 = recognition cue; H2 = visual/structural model; H3 = setup skeleton only.

Question wording is paraphrased; buttons open the original ExamSIDE source.

Local ST05 slice has 5 required rows; circular and no-two-together rows are deferred.

HOW TO USE THIS COMPANION

Attempt -> minimum helper -> diagnose -> Appendix A

MAP

Question map

Q1

2026 queue: girls not all together

C3/C5 | D2

Q2

2025 DAUGHTER vowels not all together

C3 | D2

Q3

2023 MATHEMATICS C/S apart

C4 | D4

Q4

2021 VOWELS consonants not all together

C3 | D2

Q5

2020 LETTER vowels not adjacent

C2/C4 | D3

Self-diagnosis tags

NOT_ALL_VS_NO_TWO

logical scope misread

WRONG_UNIVERSE

TOTAL/BAD mismatch

BLOCK_INTERNAL_ORDER

bad-event block undercounted

REPEAT_SYMMETRY_LOST

multiset bad count wrong

ONE_SUBTRACTION_OVERLAP

several bad events collapsed

RULE

Practice integrity

Open the smallest helper that restores the model. H2/H3 should repair the complement partition, not replace an honest first attempt.

BOUNDARY

Primary owner

If the question says no two of a whole group are adjacent, expect ST06 gaps. If it is circular, the circular unit owns the universe.

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3/C5

NOT_ALL_VS_NO_TWO

D2

Question

QUESTION

JEE Main 2026 - 4 Apr Morning

Four boys and three girls form one queue. Count the queues in which the three girls are not all together as one consecutive block.

H1

Recognition cue

Interpret the condition as NOT(all 3 girls together), not as “no two girls adjacent.”

Your working

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3

NOT_ALL_VS_NO_TWO

D2

Question

QUESTION

JEE Main 2025 - 23 Jan Morning

Using all eight distinct letters of DAUGHTER, count the arrangements in which the three vowels A, U and E are not all together in one consecutive block.

H1

Recognition cue

Count all $8!$ arrangements, then remove the arrangements in which A,U,E form one block.

Your working

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C4

REPEAT_SYMMETRY_LOST

D4

Question

QUESTION

JEE Main 2023 - 8 Apr Evening

Using all letters of MATHEMATICS, C and S must not be adjacent. If the number of valid words is $(6!)k$, find k .

H1

Recognition cue

Use complement: TOTAL multiset arrangements minus the arrangements where C and S are one adjacent block.

H2

Visual/concept helper

TOTAL inventory: 11 letters with M2,A2,T2. BAD inventory: 10 outer objects after gluing C,S; the block has 2 internal orders and M,A,T remain repeated.

H3

Structural skeleton

Skeleton: $11!/(2!2!2!) - 2 \times 10!/(2!2!2!) = (6!)k$. Stop before arithmetic.

Q4

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3

NOT_ALL_VS_NO_TWO

D2

Question

QUESTION

JEE Main 2021 - 31 Aug Morning

Using all six distinct letters of VOWELS, count the arrangements in which the four consonants V,W,L,S are not all together as one consecutive block.

H1

Recognition cue

The forbidden event is one 4-consonant block. Two or three consonants may still be adjacent in a valid word.

Your working

Q5

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C2/C4

REPEAT_SYMMETRY_LOST

D3

Question

QUESTION

JEE Main 2020 - 6 Sep Evening

Using all letters of LETTER, count the distinct arrangements in which the two vowels E,E are not adjacent.

H1

Recognition cue

Because there are exactly two vowels, “never come together” means the E,E pair must not be adjacent. Use TOTAL - adjacent-E block.

H2

Visual/concept helper

TOTAL inventory: L,E,E,T,T,R. BAD inventory: [EE],L,T,T,R. The E block has only one internal order because the E letters are identical.

Your working

DIAGNOSTIC CHECKPOINT

Name the structural error before opening Appendix A

DIAG

Q1/Q2/Q4

Did you forbid only the all-together event, rather than every pairwise adjacency?

Q3

Did you keep repeated M,A,T symmetries in BOTH TOTAL and BAD?

Q5

Did you avoid multiplying the [EE] block by 2, since the E letters are identical?

RETRY RULE

Repair before reveal

If the error is structural, return to the linked Study Guide concept and rewrite Universe / Forbidden / Good. Do not use Appendix A as the first repair.

LOCAL COVERAGE RECONCILIATION

Self-check against the frozen ST05 slice

AUDIT

Required rows

2026 queue girls not all together C3/C5 Q1 **PASS**

2025 DAUGHTER vowels not all together C3 Q2 **PASS**

2023 MATHEMATICS C/S apart C4 Q3 **PASS**

2021 VOWELS consonants not all together C3 Q4 **PASS**

2020 LETTER E,E apart C2/C4 Q5 **PASS**

Deferred rows

2017 round-table B1/G1 never adjacent **DEFER -> circular permutation**

2008 MISSISSIPPI no two S adjacent **DEFER -> ST06 gap method**

2023 round table no two girls **DEFER -> ST06 + circular**

2021 FARMER excluding RR + dictionary rank **DEFER -> dictionary-rank hybrid**

2025 B1/B2 inside boys block **SECONDARY revisit only; ST04 primary**

RECONCILE

Local ST05 status: **PASS**

Required: 5
Included: 5/5
Source links: 5/5
Concept links: 5/5
Appendix solutions: 5/5
H-policy failures: 0

CHAPTER STATUS

Audit boundary

PROVISIONAL. This local ST05 slice is reconciled, but chapter **COMPLETE** still requires every one of the 220 P&C; rows to be classified.

A1

SOLUTION

Reasoning

All 7 students are distinct, so $TOTAL = 7!$. BAD means all three girls are together: glue them into one block. Then 4 boys + 1 girls-block give $5!$ outer arrangements, and the 3 girls have $3!$ internal orders. Thus $GOOD = 7! - 5! \times 3! = 5040 - 720 = 4320$.

ANSWER

Final result

4320

A2

SOLUTION

Reasoning

DAUGHTER has 8 distinct letters. TOTAL = $8!$. BAD means A,U,E are all together. Glue the three vowels into one block: 5 consonants + 1 vowel-block give $6!$ outer orders, and the vowels have $3!$ internal orders. Therefore GOOD = $8! - 6! \times 3! = 40320 - 4320 = 36000$.

ANSWER

Final result

36000

A3

SOLUTION

Reasoning

MATHEMATICS has 11 letters with M,A,T repeated twice. $TOTAL = 11!/(2!2!2!)$. For BAD, glue C and S; the block has 2 internal orders. The 10 outer objects still contain M2,A2,T2, so $BAD = 2 \times 10!/(2!2!2!)$. $GOOD = 4,082,400$. Since $4,082,400 = 6! \times 5670$, $k = 5670$.

ANSWER

Final result

5670

A4

SOLUTION

Reasoning

VOWELS has 6 distinct letters. TOTAL = $6!$. BAD means the four consonants V,W,L,S form one consecutive block. Then that block together with O and E gives $3!$ outer orders, and the four consonants have $4!$ internal orders. GOOD = $6! - 3! \times 4! = 720 - 144 = 576$.

ANSWER

Final result

576

A5

SOLUTION

Reasoning

LETTER has multiplicities E2 and T2. $TOTAL = 6!/(2!2!) = 180$. BAD means the identical E letters are adjacent; treat EE as one block. The outer inventory [EE], L, T, T, R has 5 objects with T repeated twice, so $BAD = 5!/2! = 60$. $GOOD = 180 - 60 = 120$.

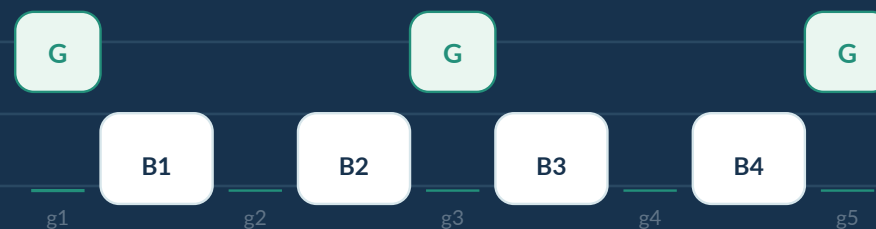
ANSWER

Final result

120

EXAMSIDE PRACTICE: GAP METHOD

Three frozen PYQs | concept backlinks | progressive H1-H3 | Appendix A solutions



PRACTICE RULE

Attempt first. Use H1 only if recognition fails; H2 only if the model is still unclear; H3 only when setup remains blocked.

Question pages never reveal final arithmetic. Complete solutions live only in Appendix A.

Keep this PDF in the same folder as the ST06 Study Guide so concept-return buttons can resolve in compatible readers.

LOCAL COVERAGE LEDGER

One primary owner per ExamSIDE question

AUDIT

REQUIRED for ST06

Q1

2025

D4

5 boys + 4 girls: OR with no-two-boys branch

ST06.C7

Q2

2022

D4

11 blue + 5 red cubes, at least two blue apart

ST06.C6

Q3

2008

D3

MISSISSIPPI, no two S adjacent

ST06.C4

RECONCILIATION

Local release gate

Required: 3 | Included: 3/3

Source links: 3/3 | Concept links: 3/3

Appendix solutions: 3/3 | H-policy failures: 0

DEFER - do not inflate ST06

CIRCULAR

Deferred owner

2023: 5 girls + 7 boys around a round table, no two girls together. Uses cyclic gaps but circular symmetry dominates -> Circular unit.

OPEN 2023 CIRCULAR PYQ

AIEEE 2003

Deferred owner

6 men + 5 women at a round table, no two women together. Same defer rule: circular anchor count and cyclic gaps must be taught together.

OPEN CHAPTER CORPUS

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO ST06.C7

FORMULA_FIRST_RESTRICTION_HIDDEN

D4

Question

QUESTION

JEE Main 2025 | 23 Jan Evening

Five distinct boys and four distinct girls sit in a row. Count the arrangements in which either all five boys sit together or no two boys sit together.

H1

Recognition cue

The word either creates two branches. Use a block for the all-boys-together branch and gaps for the no-two-boys-together branch.

H2

Visual/concept helper

For the gap branch, arrange the 4 girls first. They create exactly 5 gaps, so all 5 boys must occupy one distinct gap each.

H3

Structural skeleton

Branch A: $5! \times 5!$

Branch B: $4! \times 5!$

Check whether the two events overlap before adding.

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO ST06.C6

GAP_OFF_BY_ONE

D4

Question

QUESTION

JEE Main 2022 | 27 Jun Morning

Sixteen identical cubes consist of 11 blue cubes and 5 red cubes. They are placed in a row so that between any two red cubes there are at least two blue cubes. Find the number of arrangements.

H1

Recognition cue

Do not start by choosing red positions. First build the minimum legal red-blue scaffold.

H2

Visual/concept helper

Five red cubes create four internal red-to-red gaps. Put 2 compulsory blue cubes in each internal gap: that uses 8 of the 11 blue cubes.

H3

Structural skeleton

Three blue cubes remain. Distribute them among 6 regions (two ends + four internal regions): solve $x_1 + \dots + x_6 = 3$ with $x_i \geq 0$.

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO ST06.C4

REPETITION_ACCIDENTALLY_ALLOWED

D3

Question

QUESTION

AIEEE 2008

How many distinct words can be formed by arranging all the letters of MISSISSIPPI so that no two S letters are adjacent?

H1

Recognition cue

Remove the restricted S letters first. Arrange the remaining letters, respecting repeated I and P.

H2

Visual/concept helper

Every realized base row of M, I,I,I,I, P,P has 8 gaps. Choose 4 distinct gaps for the four identical S letters.

Your working

DIAGNOSTIC REVIEW

Name the architecture before checking Appendix A

REVIEW

For each question, write...

Q1

Hybrid control

1. Why are the two OR branches disjoint?
2. Which branch uses ST04?
3. Why do 4 girls create exactly 5 usable gaps for 5 boys?

Q3

Repeated-object transfer

1. What is the non-S multiset?
2. Why does each base row have 8 gaps despite repeated letters?
3. Why is there no 4! for the S letters?

Q2

Stronger spacing

1. How many blue cubes are compulsory?
2. Why are there 6 residual regions?
3. Why is this not simply $C(12,5)$?

ERROR TAG

Diagnose, do not just score

Mark any miss as one of:

GAP_OFF_BY_ONE / REPETITION ERROR / WRONG OWNER /
AT-LEAST-AS-EXACTLY / IDENTITY ASSIGNMENT / OR-OVERLAP

A1

Branch A: all boys together

SOLUTION

ST04 branch

Treat the five boys as one super-object. Together with the four girls there are 5 outer objects, so $5!$ outer orders. The five boys can permute internally in $5!$ ways.

$$A = 5! \times 5! = 14400.$$

Branch B: no two boys together

SOLUTION

ST06 branch

Arrange the four girls first: $4!$. Their row creates 5 gaps. Because there are exactly five distinct boys, every gap must contain one boy; assign the boys in $5!$ ways.

$$B = 4! \times 5! = 2880.$$

FINAL

Answer

The branches are disjoint: five boys cannot be all consecutive and simultaneously pairwise nonadjacent. Therefore

$$A + B = 14400 + 2880 = 17280.$$

A2

Build the minimum legal pattern

STEP 1

Compulsory scaffold

Five red cubes create four internal separations. Each needs at least two blue cubes, so reserve 8 blue cubes:

R BB R BB R BB R BB R.

Count the leftovers

STARS AND BARS

Six regions

Let $x_1, \dots, x_6$ be the numbers of extra blue cubes in the six regions. Then

$$x_1 + \dots + x_6 = 3, x_i \geq 0.$$

$$\text{Number of solutions} = C(3+6-1, 6-1) = C(8, 5).$$

STEP 2

Residual freedom

There are $11 - 8 = 3$ blue cubes left. These can be placed in any of 6 regions: before the first R, after the last R, or added to any of the four internal blue runs.

$$C(8.5) = 56$$

ANSWER

Answer

All cubes of each color are identical, so no further arrangement factor is needed.

Required number = 56.

A3

Arrange the non-S letters first

STEP 1

Multiset base

Removing the four S letters leaves M, I,I,I,I, P,P. The number of distinct base rows is

$$7!/(4!2!) = 105.$$

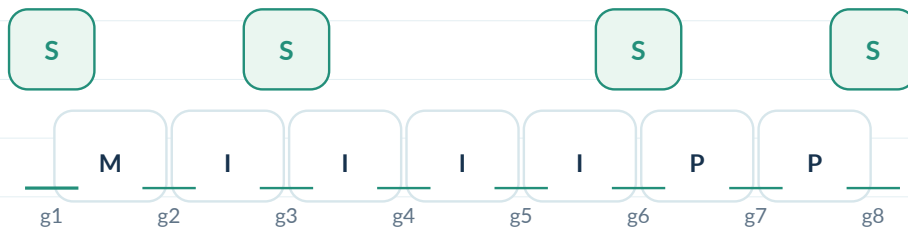
Choose safe gaps for S

STEP 2

No 4! factor

Each base row has 8 gaps. To ensure no two S letters are adjacent, place at most one S in a gap. The S letters are identical, so choose 4 of the 8 gaps:

$$C(8,4) = 70.$$



$$105 \times 70 = 7350$$

ANSWER

Answer

Required number of distinct words = 7350. This matches option D on the ExamSIDE question.

AUDIT

Practice integrity

PASS

Q1 source link + concept backlink + D4 H1/H2/H3

PASS

Q2 source link + concept backlink + D4 H1/H2/H3

PASS

Q3 source link + concept backlink + D3 H1/H2

PASS

All final answers isolated in Appendix A

PASS

No circular-gap question counted as ST06 REQUIRED

Question/helper/solution prose kept above readability floor

LOCAL RESULT

PASS

Required 3 | Included 3/3 | Source links 3/3 | Concept links 3/3 | Solutions 3/3 | H-policy failures 0 | Readability failures 0

Corpus boundary

DEFER

Named destination

2023 round-table no-two-girls and AIEEE 2003 round-table no-two-women are not missing. They are intentionally deferred to the circular-permutation subtopic because cyclic symmetry changes both the anchor and gap counts.

CHAPTER STATUS

Audit honesty

The parent ExamSIDE chapter contains 220 P&C; questions. ST06 local coverage is frozen and passes, but chapter-wide status remains PROVISIONAL until every source row has a final owner/disposition.

OPEN PARENT EXAMSIDE CORPUS

START

HOW TO USE

Helper policy

Attempt each question first. H1 restores recognition, H2 shows the structural representation, and H3 gives only the counting skeleton. Full solutions are isolated in Appendix A.

Frozen required set

Q1

2021 | even number; zero; no repetition

D3

Q2

2023 | >40000 + divisible by 5

D3

Q3

2022 | multiple of 15; no repetition

D4

Q4

2026 | interval + divisible by 3 + repetition

D4

Q5

2026 | each digit at least once

D4

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C1/C4

ZERO_LEADING

D3

Question

QUESTION

JEE Main 2021 | 26 Aug Morning

How many three-digit even numbers can be formed from 0, 1, 3, 4, 6, 7 without repeating a digit?

H1

Recognition cue

Fix the units digit first. Why is units = 0 structurally different from units = 4 or 6?

H2

Visual / concept helper

Split into two branches: last digit 0; last digit one of {4,6}. Track the hundreds-place pool in each branch.

Your working

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3/C4

ZERO_LEADING

D3

Question

QUESTION

JEE Main 2023 | 12 Apr Morning

Using 0,1,3,5,7,9 without repetition, count the five-digit numbers greater than 40000 and divisible by 5.

H1

Recognition cue

Two slots are special: the first digit controls the range; the last digit controls divisibility by 5.

H2

Visual / concept helper

Split units = 0 and units = 5. In each branch, count legal first digits before filling the middle three positions.

Your working

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C5/C6

FORMULA_FIRST_RESTRICTION_HIDDEN

D4

Question

QUESTION

JEE Main 2022 | 30 Jun Morning

How many six-digit numbers can be made from 1,2,3,4,5,6,7 without repetition and be divisible by 15?

H1

Recognition cue

Divisible by 15 means divisibility by 5 and by 3 simultaneously. Which condition fixes a position?

H2

Visual / concept helper

The units digit is forced to 5. Now choose which five of the remaining six digits make the total digit sum divisible by 3.

H3

Structural skeleton

For each legal five-digit selection: arrange those five digits in the first five positions $\rightarrow 5!$. Add the legal selection cases.

Q4

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3/C5

FORMULA_FIRST_RESTRICTION_HIDDEN

D4

Question

QUESTION

JEE Main 2026 | 24 Jan Morning

Using digits 0,1,2,5,9 with repetition allowed, count numbers greater than 5000, less than 9000 and divisible by 3.

H1

Recognition cue

The interval forces the thousands digit. Which available digit can it be?

H2

Visual / concept helper

After fixing the first digit at 5, group the allowed digits by residue mod 3:
residue 0 has {0,9}, residue 1 has {1}, residue 2 has {2,5}.

H3

Structural skeleton

Count ordered residue triples whose sum is $1 \pmod 3$, weighting each residue choice by the number of actual digits in that class.

Q5

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C8

IE_OVERLAP_MISSED

D4

Question

QUESTION

JEE Main 2026 | 2 Apr Morning

Seven-digit numbers are formed using only 1,2,3,5,7, and every one of the five digits must occur at least once. Count them.

H1

Recognition cue

Start from all 5^7 strings. The forbidden events are “digit d is missing.”
These bad events overlap.

H2

Visual / concept helper

Use inclusion-exclusion over the five missing-digit events, or classify multiplicities as 3,1,1,1,1 and 2,2,1,1,1.

H3

Structural skeleton

IE skeleton: $5^7 - C(5,1)4^7 + C(5,2)3^7 - C(5,3)2^7 + C(5,4)$.

A

ANSWER

Verified result

52

Teacher solution

SOLUTION

Decision sequence

Case 1: units digit 0. Hundreds digit has 5 nonzero choices; tens then has 4 choices, giving 20.

Case 2: units digit is 4 or 6 (2 choices). The hundreds digit cannot be 0 and cannot be the used units digit, so it has 4 choices. The tens digit then has 4 remaining choices. Count = $2 \times 4 \times 4 = 32$.

Total = $20 + 32 = 52$.

A

ANSWER

Verified result

120

Teacher solution

SOLUTION

Decision sequence

Units must be 0 or 5.

Units = 0: first digit must exceed 4, so it is 5, 7 or 9: 3 choices. Fill the middle 3 slots by ordering 3 of the remaining 4 digits: $4P3 = 24$. Count 72.

Units = 5: first digit must be 7 or 9: 2 choices. Again $4P3 = 24$ for the middle positions. Count 48.

Total = 120.

A

ANSWER

Verified result

360

Teacher solution

SOLUTION

Decision sequence

Divisible by 5 forces the last digit to be 5. The six remaining digits are 1,2,3,4,6,7; choose five so the total sum including 5 is divisible by 3.

The legal omitted digits are 7, 1 and 4, equivalently the legal five-sets are {1,2,3,4,6}, {2,3,4,6,7}, {1,2,3,6,7}. Each set can be arranged in the first five positions in 5! ways.

Total = $3 \times 5! = 360$.

A

ANSWER

Verified result

42

Teacher solution

SOLUTION

Decision sequence

The interval and digit set force the thousands digit to be 5. We therefore count triples (a,b,c) from $\{0,1,2,5,9\}$, repetition allowed, for which $a+b+c$ is congruent to 1 mod 3 because 5 is congruent to 2 mod 3.

Residue-class sizes are $r_0=2$, $r_1=1$, $r_2=2$. The ordered residue patterns summing to 1 mod 3 are $(0,0,1)$, $(0,2,2)$, $(1,1,2)$, and their permutations where distinct. Weighted counts: $3*(2*2*1) + 3*(2*2*2) + 3*(1*1*2) = 12 + 24 + 6 = 42$.

A

ANSWER

Verified result

16800

Teacher solution

SOLUTION

Decision sequence

Use inclusion-exclusion on the five events that a particular digit is absent:

$$5^7 - C(5,1)4^7 + C(5,2)3^7 - C(5,3)2^7 + C(5,4)1^7$$

$$= 78125 - 81920 + 21870 - 1280 + 5 = 16800.$$

LOCAL EXAMSIDE COVERAGE RECONCILIATION

One owner per question; no count inflation

AUDIT

Release ledger

Required 5

Included 5/5

Source links 5/5

Concept backlinks 5/5

Appendix solutions 5/5

H-policy failures 0

Pedagogy-layer failures 0

Readability failures 0

LOCAL STATUS ST07 release stamp

PASS for the frozen subtopic slice. Chapter-wide status remains PROVISIONAL until all 220 ExamSIDE P&C; rows have a frozen owner/disposition.

Deferred / boundary

D1

Dictionary-ordered numbers/words
Destination: ST08 Dictionary Rank

D2

Repeated-digit positional multiset problems
Destination: ST02 / ST03

D3

Forbidden digit strings requiring inclusion-exclusion over blocks
Destination: Later forbidden-pattern unit

AUDIT RULE

No drift

A question is never moved into this subtopic simply to increase the practice count. Hybrid questions retain one primary owner and may only be revisited explicitly.

START

HELPER POLICY

Attempt before opening helpers

H1 points to the next prefix decision. H2 shows the bucket structure. H3 gives the rank-table skeleton but never the completed arithmetic.

Q1

MONDAY

D3

Q2

THAMS

D3

Q3

PUBLIC

D3

Q4

BHBJO - 50th word

D4

Q5

UDAYPUR

D4

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C2/C4

DICTIONARY_PREFIX_MISCOUNT

D3

Question

QUESTION

JEE Main 2023 | 13 Apr Evening

All permutations of MONDAY are listed in English-dictionary order. Find the serial number of MONDAY.

H1

Recognition cue

At the first position, which available letters are alphabetically smaller than M? Each creates one complete suffix block.

H2

Visual / concept helper

After fixing M, repeat with target O using only the remaining letters. Keep a running total of completed prefix buckets.

Your working

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C2/C4

DICTIONARY_PREFIX_MISCOUNT

D3

Question

QUESTION

JEE Main 2023 | 11 Apr Evening

All permutations of MATHS are listed in dictionary order. Find the serial number of THAMS.

H1

Recognition cue

All words beginning with A,H,M,S come before every T... word.

H2

Visual / concept helper

After reaching T..., only TA... comes before TH.... Count that 3! block, then inspect the remaining suffix.

Your working

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C2/C4

OFF_BY_ONE

D3

Question

QUESTION

JEE Main 2023 | 6 Apr Evening

All permutations of PUBLIC are listed in dictionary order. Find the serial number of PUBLIC.

H1

Recognition cue

Build the rank left-to-right; every smaller unused letter at a position contributes a factorial suffix block.

H2

Visual / concept helper

Sorted letters are B,C,I,L,P,U. Start by skipping B..., C..., I..., L... before entering P...

Your working

Q4

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C5/C6

DICTIONARY_PREFIX_MISCOUNT

D4

Question

QUESTION

JEE Main 2024 | 5 Apr Evening

The 60 distinct permutations of BHBJO are written in dictionary order. Find the 50th word.

H1

Recognition cue

This is an inverse-rank problem. Locate which first-letter bucket contains rank 50.

H2

Visual / concept helper

First-letter bucket sizes are unequal: B... has 24 words, while H..., J..., O... each have 12 because both B copies remain.

H3

Structural skeleton

Ranks 1-24: B...; 25-36: H...; 37-48: J...; therefore enter O... with within-bucket rank 2, then recurse on B,B,H,J.

Q5

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C3/C4

DICTIONARY_PREFIX_MISCOUNT

D4

Question

QUESTION

JEE Main 2026 | 24 Jan Evening

All distinct permutations of UDAYPUR are arranged in dictionary order. Find the rank of UDAYPUR.

H1

Recognition cue

Repeated U means a prefix bucket may have size $6!/2!$, not $6!$.

H2

Visual / concept helper

Sorted letters: A,D,P,R,U,U,Y. Count complete A..., D..., P..., R... buckets before entering U.... Recompute the suffix multiplicity after every matched prefix.

H3

Structural skeleton

Use a row per target position: [target letter] x [smaller unused letters] x [distinct permutations of remaining suffix]. Sum all rows, then +1.

A

ANSWER

Verified result

327

Teacher solution

SOLUTION

Decision sequence

Sorted letters: A,D,M,N,O,Y.

Before M: A... and D... contribute $2 \times 5! = 240$.

Within M..., before O: A,D,N contribute $3 \times 4! = 72 \rightarrow$ total 312.

Within MO..., before N: A,D contribute $2 \times 3! = 12 \rightarrow$ 324.

Within MON..., before D: A contributes $2! = 2 \rightarrow$ 326.

The remaining target suffix is forced. Rank = $326 + 1 = 327$.

A

ANSWER

Verified result

103

Teacher solution

SOLUTION

Decision sequence

Sorted letters: A,H,M,S,T.

Before T: A,H,M,S give $4 \times 4! = 96$.

Inside T..., before H the smaller available letter A gives $3! = 6$.

After TH, target suffix AMS is the first lexicographic arrangement of A,M,S, so no more complete blocks occur.

Words before = 102; rank = 103.

A

ANSWER

Verified result

582

Teacher solution

SOLUTION

Decision sequence

Sorted letters: B,C,I,L,P,U.

Before P: B,C,I,L $\rightarrow 4 \times 5! = 480$.

Within P..., before U: B,C,I,L $\rightarrow 4 \times 4! = 96 \rightarrow 576$.

Within PU..., before B: none.

Within PUB..., before L: C,I $\rightarrow 2 \times 2! = 4 \rightarrow 580$.

Within PUBL..., before I: none; remaining suffix forced.

Rank = $580 + 1$? Continue carefully: after PUBL, target I then C. At PUB position target B, no smaller. At PUBL target L, remaining C,I,L: smaller C,I each leave $2! = 2$, so +4. Running 580. At PUBLI target I, remaining C,I: smaller C leaves $1! = 1 \rightarrow 581$. Then target C. Rank = 582.

A

ANSWER

Verified result

OBBJH

Teacher solution

SOLUTION

Decision sequence

Distinct first-letter buckets:

B...: remaining B,H,J,O $\rightarrow 4! = 24$ (ranks 1-24).

H...: remaining B,B,J,O $\rightarrow 4!/2! = 12$ (25-36).

J...: 12 (37-48).

O...: 12 (49-60).

So rank 50 is the 2nd word inside O.... Remaining letters B,B,H,J. Their ordered distinct suffixes begin BBHJ, BBJH, ... Therefore the 2nd is BBJH.

50th word = OBBJH.

A

ANSWER

Verified result

1580

Teacher solution

SOLUTION

Decision sequence

Sorted multiset A,D,P,R,U,U,Y.

Before first U: A,D,P,R each contributes $6!/2! = 360 \rightarrow 1440$.

Fix U. Target next D; smaller A contributes $5! = 120 \rightarrow 1560$.

Fix UD. Target A: none.

Fix UDA. Target Y; smaller P,R,U contribute $3!$ each $= 18 \rightarrow 1578$.

Fix UDAY. Target P: none.

Fix UDAYP. Target U; smaller R contributes $1! = 1 \rightarrow 1579$ words before.

Therefore rank = 1580.

LOCAL EXAMSIDE COVERAGE RECONCILIATION

One owner per question; no count inflation

AUDIT

Release ledger

Required 5

Included 5/5

Source links 5/5

Concept backlinks 5/5

Appendix solutions 5/5

H-policy failures 0

Pedagogy-layer failures 0

Readability failures 0

LOCAL STATUS ST08 release stamp

PASS for the frozen subtopic slice. Chapter-wide status remains PROVISIONAL until all 220 ExamSIDE P&C; rows have a frozen owner/disposition.

Deferred / boundary

D1

Alphabetical-order constraints without serial/rank
Destination: ST03 relative-order restrictions

D2

General repeated-word arrangements
Destination: ST02

D3

Probability recurrence after rank is found
Destination: Probability; ST08 remains primary owner only when rank is the key recognition

AUDIT RULE

No drift

A question is never moved into this subtopic simply to increase the practice count. Hybrid questions retain one primary owner and may only be revisited explicitly.

START

SCOPE Do not inflate the corpus

This practice pack contains the three direct circular-seating owners found in the frozen JEE Main/AIEEE ExamSIDE slice. Low question count is an audit result, not a coverage gap.

Q1

2023 | 5 girls + 7 boys; no two girls

D4

Q2

2017 | specified B1/G1 non-adjacent

D3

Q3

AIEEE 2003 | 6 men + 5 women; no two women

D3

Q1

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C6

GAP_OFF_BY_ONE

D4

Question

QUESTION

JEE Main 2023 | 8 Apr Morning

In how many ways can 5 girls and 7 boys be seated at a round table so that no two girls sit together?

H1

Recognition cue

Arrange the separator group (boys) around the circle first. How many cyclic gaps do 7 boys create?

H2

Visual / concept helper

Seven boys give 7 cyclic gaps, not 8. Choose 5 gaps and assign the 5 distinct girls to them.

H3

Structural skeleton

Skeleton: $6! \times C(7,5) \times 5!$.

Q2

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C5

CIRCULAR_ROTATION_OVERCOUNT

D3

Question

QUESTION

JEE Main 2017 | 9 Apr Morning

Five boys and three girls sit at a round table. A specified boy B1 and specified girl G1 must not be adjacent. Count the arrangements.

H1

Recognition cue

Use complement: total circular seatings minus seatings with B1 and G1 glued together.

H2

Visual / concept helper

Total for 8 people is $7!$. In the bad case, the pair becomes one unit: 7 circular units, with 2 internal orders.

Your working

Q3

OPEN ORIGINAL EXAMSIDE QUESTION

BACK TO C6/C7

GAP_OFF_BY_ONE

D3

Question

QUESTION

AIEEE 2003

Six men and five women dine at a round table with no two women together. Count the arrangements.

H1

Recognition cue

Arrange the 6 men circularly first; they are the separator group.

H2

Visual / concept helper

Six men create exactly 6 cyclic gaps. Choose 5 of those gaps and arrange the 5 women.

Your working

A

ANSWER

Verified result

1,814,400

Teacher solution

SOLUTION

Decision sequence

Arrange the 7 distinct boys around the round table: $(7-1)! = 6!$.

These boys create 7 cyclic gaps. To keep the 5 girls separated, choose 5 of those 7 gaps: $C(7,5)$. Arrange the 5 distinct girls in the selected gaps: $5!$.

Total = $6! \times C(7,5) \times 5! = 1,814,400$.

A

ANSWER

Verified result

3600

Teacher solution

SOLUTION

Decision sequence

Total circular arrangements of 8 distinct people = $7!$.

Bad arrangements: treat B1 and G1 as a block. The block plus the other 6 people gives 7 circular units, arranged in $(7-1)! = 6!$ ways. The block has 2 internal orders.

Good = $7! - 2 \times 6! = 3600 = 5 \times 6!$.

A

ANSWER

Verified result

 $6! \times 5!$

Teacher solution

SOLUTION

Decision sequence

Arrange the 6 men circularly: $5!$. They create 6 cyclic gaps. Choose 5 of the 6 gaps for the women: $C(6,5)=6$. Arrange the 5 distinct women: $5!$.

Total = $5! \times 6 \times 5! = 6! \times 5!$.

LOCAL EXAMSIDE COVERAGE RECONCILIATION

One owner per question; no count inflation

AUDIT

Release ledger

Required 3

Included 3/3

Source links 3/3

Concept backlinks 3/3

Appendix solutions 3/3

H-policy failures 0

Pedagogy-layer failures 0

Readability failures 0

LOCAL STATUS ST09 release stamp

PASS for the frozen subtopic slice. Chapter-wide status remains PROVISIONAL until all 220 ExamSIDE P&C; rows have a frozen owner/disposition.

Deferred / boundary

D1 Linear no-two-together
Destination: ST06 Gap Method

D2 Linear specified pair not adjacent
Destination: ST05 Complement

D3 Circle geometry questions about chords/triangles
Destination: EXCLUDE - not permutation seating

AUDIT RULE No drift

A question is never moved into this subtopic simply to increase the practice count. Hybrid questions retain one primary owner and may only be revisited explicitly.

DERANGEMENTS & FORBIDDEN POSITIONS - EXAMSIDE PRACTICE

Source: [link](#) to questions + progressive helpers

METHOD

How to use this companion

Try the question first. Open H1 only if recognition fails. H2 gives a visual/concept model. H3 gives the counting skeleton but stops before the final arithmetic. Full solutions are in Appendix A.

AUDIT

Coverage rule

One direct JEE Main derangement owner is frozen. Three ExamSIDE JEE Advanced/legacy extensions are included for depth and are explicitly excluded from the JEE Main coverage numerator.

H-POLICY

Helper policy

D1-D2: usually no helper or H1 only.

D3: H1 + H2.

D4-D5: H1 + H2 + H3. A helper must reveal structure, not merely restate the question.

QUESTION MAP

Each question is assigned to one primary concept owner

Q1 | JEE Main 2023 - 11 Apr AM | REQUIRED | D2

5 students; nobody in allotted seat

Owner: C4 Misconception: Treat wrong seats as independent choices

Q2 | JEE Advanced 2014 - Paper 2 | EXTENSION | D4

6 cards/envelopes; none matching; card 1 forced to envelope 2

Owner: C7 Misconception: Call the residual problem D_5 without re-auditing restrictions

Q3 | IIT-JEE 1988 | EXTENSION | D2

4 coloured balls into matching-colour boxes; none match

Owner: C3 Misconception: Subtract only four bad placements

Q4 | JEE Advanced 2018 - Paper 1 | EXTENSION | D3

S1 fixed; remaining four all wrong

Owner: C6 Misconception: Forget that the remaining students must be deranged

D2

Open original ExamSIDE question

Study Guide: C4 - page 5

PYQ

Question (paraphrased)

Five distinct students are assigned five labelled seats according to roll number. Count the seatings in which no student occupies the assigned seat.

H1

H1 - Recognition cue

Translate directly to D₅.

H2

H2 - Visual / concept helper

Think of a 5x5 assignment grid with the diagonal forbidden.

Working space: identify the structure before calculating.

Q2 | JEE Advanced 2014 - Paper 2

6 cards/envelopes; none matching; card 1 forced to envelope 2

D4

Open original ExamSIDE question

Study Guide: C7 - page 8

PYQ

Question (paraphrased)

Six numbered cards are placed one per six same-numbered envelopes. No card may go to its matching envelope, and card 1 is forced into envelope 2. Count the placements.

H1

H1 - Cue

Remove card 1 and envelope 2 first.

H2

H2 - Model

In the remaining 5×5 grid, only cards 3,4,5,6 still have diagonal bans; card 2 has no surviving own envelope.

H3

H3 - Skeleton

Use inclusion-exclusion on the four surviving fixed-point events: $5! - C(4,1)4! + C(4,2)3! - C(4,3)2! + C(4,4)$.

Working space: identify the structure before calculating.

Q3 | IIT-JEE 1988

4 coloured balls into matching-colour boxes; none match

D2

[Open original ExamSIDE question](#)

[Study Guide: C3 - page 4](#)

PYQ

Question (paraphrased)

Four distinct coloured balls are put one each into four same-coloured boxes. Count assignments in which no ball goes to the box of its own colour.

H1

H1 - Recognition cue

This is D₄.

H2

H2 - Visual / concept helper

Use small derangement values or inclusion-exclusion.

Working space: identify the structure before calculating.

D3

Open original ExamSIDE question

Study Guide: C6 - page 7

PYQ

Question (paraphrased)

Five students are randomly assigned five seats. Find the probability that S1 gets the originally assigned seat while none of S2-S5 gets the originally assigned seat.

H1

H1 - Recognition cue

Fix S1 correctly; the other four must form a derangement.

H2

H2 - Visual / concept helper

Favourable outcomes = D_4 ; total outcomes = $5!$.

Working space: identify the structure before calculating.

APPENDIX A - SOLUTIONS

Solutions are separate so helpers can remain genuinely progressive

A

APPENDIX

Solution index

Each solution begins on a fresh page. For extension questions, the label is retained so the local JEE Main coverage audit is not inflated.

A

SOLUTION

Reasoning

This is a derangement of 5 objects. Using $D_n = (n-1)(D_{n-1} + D_{n-2})$, $D_5 = 4(9+2) = 44$. Hence the number of seatings is 44.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

After fixing card 1 $\rightarrow$ envelope 2, arrange cards 2,3,4,5,6 in envelopes 1,3,4,5,6. Only cards 3-6 retain forbidden same-numbered envelopes. Inclusion-exclusion gives $5! - 4 \cdot 4! + C(4,2)3! - C(4,3)2! + 1 = 120 - 96 + 36 - 8 + 1 = 53$.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

The condition is a derangement of 4 objects. $D_4 = 4! - C(4,1)3! + C(4,2)2! - C(4,3)1! + 1 = 24 - 24 + 12 - 4 + 1 = 9$.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

Fixing S1 leaves four students who must all avoid their own seats, giving $D_4=9$ favourable permutations. Total seatings are $5!=120$.
Probability = $9/120 = 3/40$.

[Open original ExamSIDE question](#)

LOCAL COVERAGE RECONCILIATION

Self-check before the next subtopic is released

AUDIT

PASS

Corpus coverage

REQUIRED JEE Main owners frozen: 1
Included: 1/1
ExamSIDE extensions: 3
Source links: 4/4
Appendix solutions: 4/4

STAMP

Release gates

Coverage: PASS locally
Pedagogy: PASS
H-policy: PASS
Readability: PASS

Parent 220-question ledger: PROVISIONAL until fully frozen.

OWNER RULE

Boundary discipline

One direct JEE Main derangement owner is frozen. Three ExamSIDE JEE Advanced/legacy extensions are included for depth and are explicitly excluded from the JEE Main coverage numerator.

FORBIDDEN STRINGS & INCLUSION-EXCLUSION - EXAMSIDE PRACTICE

source-linked questions | progressive helpers

METHOD

How to use this companion

Try the question first. Open H1 only if recognition fails. H2 gives a visual/concept model. H3 gives the counting skeleton but stops before the final arithmetic. Full solutions are in Appendix A.

AUDIT

Coverage rule

One clean direct JEE Main owner is frozen. The 2023 repeated-symbol AP-window item is intentionally deferred because the displayed source shortcut requires an overlap audit; two JEE Advanced no-00 extensions reinforce pattern avoidance without inflating JEE Main coverage.

H-POLICY

Helper policy

D1-D2: usually no helper or H1 only.

D3: H1 + H2.

D4-D5: H1 + H2 + H3. A helper must reveal structure, not merely restate the question.

QUESTION MAP

Each question is assigned to one primary concept owner

Q1 | JEE Main 2023 - 10 Apr AM | REQUIRED | D3

Permute 1,...,7 avoiding strings 153 and 2467

Owner: C4 Misconception: Subtract two bad counts without restoring overlap

Q2 | IIT-JEE 2012 - Paper 2 | EXTENSION | D3

Binary n-digit numbers with no consecutive zeros: compute b_6

Owner: C7 Misconception: Treat each zero position independently

Q3 | IIT-JEE 2012 - Paper 2 | EXTENSION | D3

Binary no-00 recurrence statement

Owner: C7 Misconception: Miss the suffix-state recurrence

D3

[Open original ExamSIDE question](#)

[Study Guide: C4 - page 5](#)

PYQ

Question (paraphrased)

Count permutations of digits 1,2,...,7 that contain neither the exact consecutive string 153 nor the exact consecutive string 2467.

H1

H1 - Cue

Let A = contains 153 and B = contains 2467.
Use Total - |A union B|.

H2

H2 - Model

$|A|=5!$, $|B|=4!$. For A intersection B, the two rigid blocks together use all 7 digits.

H3

H3 - Skeleton

Compute $7! - 5! - 4! + 2!$.

Working space: identify the structure before calculating.

D3

[Open original ExamSIDE question](#)

[Study Guide: C7 - page 8](#)

PYQ

Question (paraphrased)

Let b_n count valid n -digit binary positive integers ending in 1, where no two zeros are consecutive. Determine b_6 .

H1

H1 - Cue

A 6-digit positive binary number starts with 1 and here ends with 1.

H2

H2 - Model

Count length-4 middle binary strings with no consecutive zeros.

H3

H3 - Skeleton

Use the Fibonacci recurrence for valid binary strings or list by number of zeros.

Working space: identify the structure before calculating.

D3

[Open original ExamSIDE question](#)[Study Guide: C7 - page 8](#)

PYQ

Question (paraphrased)

For the same no-consecutive-zero binary numbers, identify the correct recurrence relation among the stated options.

H1

H1 - Cue

Split valid length- n strings by final digit.

H2

H2 - Model

Appending 1 to any valid length- $(n-1)$ string is allowed; appending 0 requires the previous digit to be 1.

H3

H3 - Skeleton

Thus $a_n = a_{n-1} + a_{n-2}$.

Working space: identify the structure before calculating.

APPENDIX A - SOLUTIONS

Solutions are separate so helpers can remain genuinely progressive

A

APPENDIX

Solution index

Each solution begins on a fresh page. For extension questions, the label is retained so the local JEE Main coverage audit is not inflated.

A

SOLUTION

Reasoning

Total permutations = $7! = 5040$. Containing 153: treat [153] as one unit with 2,4,6,7, so $5! = 120$. Containing 2467: $4! = 24$. Containing both: two rigid blocks [153] and [2467], so $2! = 2$. Required = $5040 - 120 - 24 + 2 = 4898$.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

The first and last digits are 1. The four middle positions form any length-4 binary string with no 00. Their count satisfies $f_n = f_{n-1} + f_{n-2}$, with $f_1 = 2$, $f_2 = 3$, so $f_4 = 8$. Hence $b_6 = 8$.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

Let $a_n = b_n + c_n$, where b_n ends in 1 and c_n ends in 0. Then $b_n = a_{n-1}$, while $c_n = b_{n-1} = a_{n-2}$. Hence $a_n = a_{n-1} + a_{n-2}$. Therefore the option asserting $a_{17} = a_{16} + a_{15}$ is correct.

[Open original ExamSIDE question](#)

LOCAL COVERAGE RECONCILIATION

Self-check before the next subtopic is released

AUDIT

PASS

Corpus coverage

REQUIRED JEE Main owners frozen: 1
Included: 1/1
ExamSIDE extensions: 2
Source links: 3/3
Appendix solutions: 3/3

STAMP

Release gates

Coverage: PASS locally
Pedagogy: PASS
H-policy: PASS
Readability: PASS

Parent 220-question ledger: PROVISIONAL until fully frozen.

OWNER RULE

Boundary discipline

One clean direct JEE Main owner is frozen. The 2023 repeated-symbol AP-window item is intentionally deferred because the displayed source shortcut requires an overlap audit; two JEE Advanced no-00 extensions reinforce pattern avoidance without inflating JEE Main coverage.

HYBRID PERMUTATION STRUCTURES - EXAMSIDE PRACTICE

Source-linked questions + progressive helpers

METHOD

How to use this companion

Try the question first. Open H1 only if recognition fails. H2 gives a visual/concept model. H3 gives the counting skeleton but stops before the final arithmetic. Full solutions are in Appendix A.

AUDIT

Coverage rule

Four JEE Main permutation-enabled hybrids are frozen. Pure committee, stars-and-bars, and nCr -only problems remain excluded from the permutation-only corpus.

H-POLICY

Helper policy

D1-D2: usually no helper or H1 only.

D3: H1 + H2.

D4-D5: H1 + H2 + H3. A helper must reveal structure, not merely restate the question.

QUESTION MAP

Each question is assigned to one primary concept owner

Q1 | JEE Main 2026 - 6 Apr AM | REQUIRED | D3

INCONSEQUENTIAL: exactly 2 vowels + 2 consonants, no repeated letter

Owner: C2 Misconception: Count repeated source occurrences as distinct letters

Q2 | JEE Main 2026 - 6 Apr PM | REQUIRED | D3

9 lift passengers split 4+5, different permitted floors

Owner: C3 Misconception: Choose two floors without assigning them to group sizes

Q3 | JEE Main 2025 - 29 Jan AM | REQUIRED | D4

6-letter words from MATHS; every used letter appears at least twice

Owner: C4 Misconception: Ignore multiplicity partitions of 6

Q4 | JEE Main 2023 - 6 Apr PM | REQUIRED | D2

UNIVERSE: 2 vowels + 2 consonants, no repetition

Owner: C2 Misconception: Arrange categories before choosing symbols

D3

[Open original ExamSIDE question](#)

[Study Guide: C2 - page 3](#)

PYQ

Question (paraphrased)

Form 4-letter words from the letters appearing in INCONSEQUENTIAL, using exactly two vowels and two consonants, with no letter repeated. Count the words.

H1

H1 - Cue

First inventory distinct vowel and consonant symbols.

H2

H2 - Model

There are 5 distinct vowels and 6 distinct consonants. Choose 2 of each, then arrange the 4 selected letters.

H3

H3 - Skeleton

$C(5,2) C(6,2) 4!$.

Working space: identify the structure before calculating.

D3

[Open original ExamSIDE question](#)

[Study Guide: C3 - page 4](#)

PYQ

Question (paraphrased)

Nine people enter a lift. Exactly four exit together at one permitted floor and the remaining five at a different permitted floor. Floors 1 and 2 are skipped; floors 3-10 are available. Count the outcomes.

H1

H1 - Cue

Choose the 4-person group; the 5-person group is determined.

H2

H2 - Model

There are 8 permitted floors. Assign two distinct labelled destinations to the size-4 and size-5 groups.

H3

H3 - Skeleton

$C(9,4) \times 8P2$.

Working space: identify the structure before calculating.

D4

[Open original ExamSIDE question](#)

[Study Guide: C4 - page 5](#)

PYQ

Question (paraphrased)

Using the five symbols M,A,T,H,S with repetition allowed, count 6-letter words such that every symbol that appears occurs at least twice.

H1

H1 - Cue

The number of distinct symbols used can only be 1, 2, or 3.

H2

H2 - Model

For two symbols, allowed count splits are 2+4, 3+3, 4+2. For three symbols, only 2+2+2.

H3

H3 - Skeleton

Total = $5 + C(5,2)[\frac{6!}{(2!4!)} + \frac{6!}{(3!3!)} + \frac{6!}{(4!2!)}]$
+ $C(5,3)\frac{6!}{(2!2!2!)}.$

Working space: identify the structure before calculating.

D2

Open original ExamSIDE question

Study Guide: C2 - page 3

PYQ

Question (paraphrased)

From the letters of UNIVERSE, form 4-letter words containing exactly 2 vowels and 2 consonants, with no letter repeated. Count them.

H1

H1 - Recognition cue

Distinct vowels are U,I,E; distinct consonants are N,V,R,S.

H2

H2 - Visual / concept helper

Choose 2 vowels and 2 consonants, then arrange 4 distinct letters.

Working space: identify the structure before calculating.

APPENDIX A - SOLUTIONS

Solutions are separate so helpers can remain genuinely progressive

A

APPENDIX

Solution index

Each solution begins on a fresh page. For extension questions, the label is retained so the local JEE Main coverage audit is not inflated.

A

SOLUTION

Reasoning

Distinct vowels are I, O, E, U, A (5) and distinct consonants are N, C, S, Q, T, L (6). Choose 2 vowels and 2 consonants, then arrange the 4 chosen distinct letters: $C(5,2)C(6,2)4! = 10 \cdot 15 \cdot 24 = 3600$.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

Choose which four people form the size-4 group: $C(9,4)=126$. The two differently-sized groups must be assigned to two distinct floors among 8 permitted floors, giving $8P2=8*7=56$. Total $=126*56=7056$.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

Case 1: one symbol used $\rightarrow$ 5 words. Case 2: choose 2 symbols: $C(5,2)=10$; for each pair the counts can be (2,4),(3,3),(4,2), contributing $15+20+15=50$, hence 500. Case 3: choose 3 symbols and use each twice: $C(5,3)*6!/(2!^3)=10*90=900$. Total $=5+500+900=1405$.

[Open original ExamSIDE question](#)

A

SOLUTION

Reasoning

Distinct vowels: U,I,E (3). Distinct consonants: N,V,R,S (4). Count = $C(3,2)C(4,2)4! = 3 \cdot 6 \cdot 24 = 432$.

[Open original ExamSIDE question](#)

LOCAL COVERAGE RECONCILIATION

Self-check before the next subtopic is released

AUDIT

PASS

Corpus coverage

REQUIRED JEE Main owners frozen: 4
Included: 4/4
ExamSIDE extensions: 0
Source links: 4/4
Appendix solutions: 4/4

STAMP

Release gates

Coverage: PASS locally
Pedagogy: PASS
H-policy: PASS
Readability: PASS

Parent 220-question ledger: PROVISIONAL until fully frozen.

OWNER RULE

Boundary discipline

Four JEE Main permutation-enabled hybrids are frozen. Pure committee, stars-and-bars, and nCr-only problems remain excluded from the permutation-only corpus.